

THE STRUCTURE OF MATHEMATICAL REALITY

Phase I — Chapter 3

The Map of Mathematical Fields

Parallel Reading

Self-Study Curriculum

Parallel Reading: PCM, Part IV (“Branches of Mathematics”), read the introductions to any four branches that interest you, not the full articles, just the opening pages that state what the field studies and why. Suggested: IV.1 (Algebraic Numbers), IV.2 (Analytic Number Theory), IV.4 (Algebraic Geometry), IV.14 (Dynamics).

Also read: Saunders Mac Lane, *Mathematics: Form and Function* (Springer, 1986), Chapter 1 (“Origins of Formal Structure”), pp. 1-34. If this book is unavailable, substitute: PCM, Part I, section I.4 (“The General Goals of Mathematical Research”).